

Adapting PII Detection to Clinical Text: Utility Gains, Domain Defense, and Robustness Tradeoffs

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1. Introduction

This project investigates adversarial evasion of [openai/privacy-filter](#), a 1.5B-parameter sparse mixture-of-experts (MoE) PII classifier, focusing on two dimensions: (1) how distribution shift from general-domain to clinical text degrades detection, and (2) whether LoRA fine-tuning on clinical prose closes the adversarial gap. When PII evades detection, the consequences are concrete: patient names, dates of birth, and account numbers may end up in datasets shared with third parties under the assumption that they have been sanitized. If a model was trained predominantly on general-domain text, its behavior on clinical prose is not guaranteed, and that gap creates an exploitable attack surface.

The core finding is an attack-type-specific tradeoff. LoRA fine-tuning improves clinical recall and preserves general-domain utility. Against domain-level attacks (clinical abbreviations, honorifics, structured field formatting), fine-tuning offers no consistent defense (evasion rates shift by only ± 0.5 – 4.5 pp). Against character-level attacks (Unicode homoglyphs, invisible character injection), fine-tuning substantially increases evasion rates, scaling monotonically with training data size. We term this **adversarial robustness forgetting**: fine-tuning on clean in-domain data degrades the adversarial invariance the model previously exhibited, a failure mode invisible to standard utility metrics.

Contributions:

1. A six-attack adversarial evasion evaluation suite spanning Unicode character manipulation and clinical-domain formatting, run against both a baseline and three LoRA-adapted configurations of [openai/privacy-filter](#).
2. A PII-injected MIMIC-IV-Note evaluation set constructed by inserting AI4Privacy entities into clinical discharge summaries, enabling ground-truth F1 measurement on authentic clinical text.
3. A tradeoff analysis demonstrating that attention-only LoRA (rank 8) improves clinical F1 by +7.6 pp and preserves general-domain utility, while producing near-zero change on domain-level attack

evasion (± 0.5 – 4.5 pp) but substantially increasing character-level evasion rates (up to $+25.6$ pp for ZWSP injection). We characterize this attack-type-specific degradation as **adversarial robustness forgetting**, a distinct failure mode from catastrophic forgetting.

2. Background and Related Work

2.1 PII Detection in NLP

PII detection is commonly framed as a named entity recognition (NER) task: a token classifier assigns entity-type labels to spans of text. Modern approaches use transformer encoders with BIOES or BIO span tagging and a Viterbi decoder to enforce label consistency constraints. These models have achieved high accuracy on benchmark datasets drawn from financial and social media text, but clinical NLP presents distinct challenges: abbreviations, structured field formatting (e.g., "Attending: J. Williams, MD"), placeholder dates inserted by de-identification pipelines, and a high density of domain-specific patterns not found in general-domain corpora.

2.2 Distribution Shift as an Attack Surface

Domain shift in NLP models is well-documented: a model trained on one genre or domain degrades on another. In the context of adversarial robustness, distribution shift creates a more targeted vulnerability. If a model already handles certain surface forms poorly, such as clinical abbreviations and structured clinical fields, an adversary can exploit this by framing PII in forms the model is less likely to detect, without needing any access to the model's weights or gradients.

This project distinguishes two attack categories motivated by their interaction with domain shift: *domain-level attacks* (clinical abbreviation, honorific prepending, structured field embedding) exploit the model's unfamiliarity with clinical prose and should, in principle, be neutralized by clinical fine-tuning; *character-level attacks* (Unicode homoglyphs, zero-width space injection, thin-space injection) operate below the tokenizer and should be unaffected by clinical adaptation since they produce different byte sequences regardless of domain.

2.3 Adversarial Attacks on NLP Systems

Boucher et al. (2022) "Bad Characters: Imperceptible NLP Attacks" demonstrate that invisible or visually identical Unicode characters (homoglyphs, zero-width spaces, bidirectional control codes) that reliably fool transformer-based NLP classifiers while remaining invisible to human reviewers and requiring no model access. The paper evaluates sequence-level classifiers (sentiment, toxicity). Adapting the attack to span-level PII detection requires redefining success: a perturbation succeeds if a previously detected entity is no longer detected after perturbation. This definition is implemented in [src/adversarial_attacks.py](#).

The critical property of character-level attacks is their sub-tokenizer nature. A Cyrillic "■" substituted for a Latin "a" in a patient name produces a different token ID sequence. The model's encoder sees a different input regardless of what it has been fine-tuned on. This property implies that no amount of domain-adaptive fine-tuning on authentic clinical text can teach the model to handle Cyrillic-substituted names, because no such examples exist in any training set.

2.4 Parameter-Efficient Fine-Tuning via LoRA

Hu et al. (2022) "LoRA: Low-Rank Adaptation of Large Language Models" proposes freezing pre-trained weights and injecting trainable low-rank matrices into attention projections. With rank $r \ll d$, the number of trainable parameters is reduced from $O(d^2)$ to $O(rd)$, making adaptation of large models feasible on limited hardware. At inference, the LoRA matrices are merged with frozen weights at zero latency overhead.

For a 1.5B MoE model on a single GPU allocation, LoRA is the only tractable fine-tuning strategy. The additional complexity here is the MoE architecture: standard PEFT targets dense attention and MLP layers, but with MoE, adapting the expert weights may yield richer domain adaptation at the risk of router instability. This project uses attention-only LoRA (targeting `q_proj` and `v_proj`) as the baseline strategy, which avoids the router.

3. Approach

3.1 Target System

The victim model is `openai/privacy-filter` (HuggingFace, Apache 2.0), a bidirectional token classifier using BIOES span tagging with a constrained Viterbi decoder. Its feed-forward layers implement a sparse MoE architecture (128 experts, top-4 routing), yielding 1.5B total parameters with only ~50M active per inference. The model detects 8 entity types: `account_number`, `private_address`, `private_email`, `private_person`, `private_phone`, `private_url`, `private_date`, and `secret`. A 128k-token context window makes it suitable for long clinical notes. The model was designed as a drop-in sanitization layer for high-throughput pipelines and was not trained specifically on clinical prose.

3.2 Datasets

AI4Privacy (HuggingFace, public). 400k+ annotated examples across 50+ PII entity types drawn from financial, legal, and social media text. The 50+ fine-grained labels are mapped to the model's 8 output categories (e.g., `GIVENNAME`, `SURNAME` → `private_person`; `DATEOFBIRTH` → `private_date`; `ACCOUNTNUM` → `account_number`). Character-offset span annotations are provided in the `privacy_mask` field. This corpus serves as the general-domain benchmark throughout.

MIMIC-IV-Note 2.2 (PhysioNet, restricted). 330k+ de-identified clinical notes from Beth Israel Deaconess Medical Center, including discharge summaries, radiology reports, and progress notes. All data is stored on Hyak HPC under PhysioNet Data Use Agreement and is not uploaded to any external service. Because MIMIC is already de-identified, it cannot be used directly for F1 evaluation. It provides two things: (1) authentic clinical prose for domain characterization and LoRA fine-tuning, and (2) a text corpus into which AI4Privacy PII entities can be injected to create a labeled clinical benchmark.

PII-injected MIMIC evaluation set. Discharge summaries contain de-identification placeholders for names, dates, and other PII (e.g., ____, **[**Name**]**). A preprocessing script (`src/build_clinical_eval.py`) detects these placeholders by surrounding context patterns (e.g., `Name:`, `Attending:`, `Dr.`, `Date of Birth:`) and replaces them with AI4Privacy entities of the appropriate type, using the corresponding gold annotations as evaluation labels. The result is 22,000 labeled clinical notes. The test split uses notes 21,000–22,000 (1,000 notes held out); the training split uses notes 0–21,000. No temporal ordering is assumed; the index-based split is treated as a random partition. The test split contains 8,105 gold PII spans: 4,138 `private_person`, 3,000 `private_date`, 967 `account_number`.

3.3 Adversarial Attack Suite

Six evasion attacks are implemented in `src/adversarial_attacks.py`. Each attack takes a gold PII span, applies a perturbation, and measures whether the model still detects the span (evasion = no longer detected). All evaluations use 2,000 AI4Privacy validation examples.

Character-level attacks (expected to persist post fine-tuning):

- **Homoglyph:** substitutes Latin characters with visually identical Cyrillic Unicode characters (e.g., Latin "a" → Cyrillic "а", U+0430). Applied to `private_person`, `private_address`, `private_email`. Numeric and date-format types are excluded.
- **Zero-width space (ZWSP):** injects a zero-width space (U+200B) after the first character of each PII span. Applied to all entity types.
- **Thin-space (whitespace):** injects a Unicode thin space (U+2009) between every character of a PII span. Applied to multi-word entities (`private_person`, `private_address`). Structured formats such as email addresses are excluded on naturalism grounds.

Domain-level attacks (expected to be neutralized by clinical fine-tuning):

- **Abbreviation:** formats person names in clinical style (e.g., "John Williams" → "J. Williams"). Applicable only to `private_person`.
- **Honorific:** prepends clinical honorifics ("Dr.", "Pt.", "Attending:") to person names. Applicable only to `private_person`.

- **Field format:** embeds PII in structured clinical field patterns (e.g., "Patient: <name>", "Address: <addr>"). Applicable to `private_person` and `private_address`.

3.4 LoRA Fine-Tuning

LoRA is applied to the `q_proj` and `v_proj` attention projections only, with rank $r = 8$. Fine-tuning uses the training split of the PII-injected MIMIC dataset with training set sizes of 1k, 5k, and 20k notes. Three independent adapters are trained (`lora_r8_n1000`, `lora_r8_n5000`, `lora_r8_n20000`), enabling ablation over data quantity. Each is trained for 3 epochs with a cosine learning rate schedule. The base model weights are frozen throughout; only the LoRA matrices (~2M parameters each) are updated.

A `ForgettingCallback` evaluates general-domain F1 on a held-out AI4Privacy split at each epoch checkpoint, monitoring for catastrophic forgetting. Post-hoc corrected forgetting evaluation (`src/eval_forgetting.py`) re-evaluates saved adapters with the corrected label map and overlap matching to produce accurate utility measurements.

3.5 Evaluation Metrics

Span-level overlap F1 is the primary metric for both general-domain and clinical evaluation. The `transformers` pipeline reconstructs character-offset spans from the tokenizer's `offset_mapping` using `aggregation_strategy='simple'`. Because BPE tokenization encodes span-initial words with a leading-space prefix (e.g., `_John` rather than `John`), the reported start offset is the position of the preceding space rather than the first character of the entity. For example, a gold annotation for `John Smith` at offsets `(9, 19)` in `Patient: John Smith` will typically be predicted as `(8, 19)`. This one-character shift is systematic and is an artifact of tokenizer boundary reconstruction, not a detection failure. Overlap matching counts a prediction as a true positive if it overlaps at least one character with a gold span of the matching type, and correctly handles this misalignment. Overlap F1 is computed per document to avoid cross-document span leakage. Exact span F1 (requiring identical start and end offsets) is not reported: the systematic one-character tokenizer offset described above would artificially deflate exact match scores to near zero across all configurations, making the metric uninformative rather than diagnostic. Overlap F1 is therefore the sole span-level metric used throughout.

Evasion rate for attack evaluation is defined as: number of originally-detected spans that are no longer detected after perturbation, divided by the number of originally-detected spans. Weighted averages across entity types are computed by pooling all detected spans.

4. Evaluation

4.1 Baseline Characterization

General domain (AI4Privacy, 5,000 validation examples, overlap matching).

Entity Type	Precision	Recall	F1	Support
private_person	0.440	0.842	0.578	4,094
private_address	0.369	0.551	0.442	3,212
private_email	0.487	0.993	0.654	942
private_phone	0.351	0.964	0.515	841
private_date	0.193	0.958	0.321	620
account_number	0.080	0.914	0.146	783
secret	0.298	0.146	0.196	2,212
micro avg	0.280	0.677	0.396	12,704

The baseline shows a strong recall-precision asymmetry: the model fires broadly ($R = 0.677$) but with low precision ($P = 0.280$). This is consistent with a model designed for sanitization pipelines where missed detections are more costly than false positives. Performance is weakest on `account_number` ($F1 = 0.146$) and `secret` ($F1 = 0.196$).

Clinical domain (MIMIC-IV-Note, 500 discharge summaries, no gold labels).

Metric	Value
Notes evaluated	500
Entities detected	532
Entities per note	1.06
private_person	310 (58.3%)
private_date	207 (38.9%)
account_number	13 (2.4%)
private_address	2 (0.4%)

The model detects only 1.06 entities per discharge note. Unredacted clinical notes typically contain 5–15 PII spans. The near-zero detection of `private_address` and the dominance of `private_date` (likely firing on placeholder dates like "2180-05-07" inserted by MIMIC's de-identification pipeline) confirms that the model's behavior on clinical text is qualitatively different from general-domain.

4.2 Adversarial Attacks on Baseline Model

Evasion rates on 2,000 AI4Privacy validation examples, per attack and entity type. Not all attacks apply to all entity types; see Section 3.3 for coverage rationale.

Character-level attacks:

Attack	Entity Type	Evasion Rate
homoglyph	private_address	18.2%
homoglyph	private_email	11.4%
homoglyph	private_person	6.5%
zwsp	private_phone	72.3%
zwsp	account_number	66.9%
zwsp	private_email	66.1%
zwsp	secret	51.4%
zwsp	private_address	49.9%
zwsp	private_person	16.2%
whitespace	private_address	87.9%
whitespace	private_person	71.7%

Domain-level attacks:

Attack	Entity Type	Evasion Rate
field_format	private_address	25.3%
field_format	private_person	9.6%
abbreviation	private_person	10.2%
honorific	private_person	9.1%

The thin-space injection (whitespace) attack is the most effective, achieving 87.9% evasion on `private_address` and 71.7% on `private_person`. ZWSP injection is similarly effective on most entity types, particularly numeric formats (`private_phone` 72.3%, `account_number` 66.9%) where a single invisible character disrupts the token sequence pattern the model has learned. Homoglyph substitution is weaker (6.5–18.2%), likely because some Cyrillic–Latin pairs share the same token in the model's vocabulary. Domain-level attacks are surprisingly weak on the baseline (9.1–25.3%),

suggesting the model already has some exposure to clinical-adjacent formatting patterns during pretraining.

4.3 Post-Fine-Tuning Adversarial Robustness

Three LoRA adapters were trained at rank 8 with training set sizes of 1k, 5k, and 20k MIMIC discharge notes (Section 4.4 provides training details and utility measurements). The central result of this project is summarized below. Evasion rates are weighted averages across entity types on 2,000 AI4Privacy examples.

Attack	Type	Baseline	LoRA 1k	LoRA 5k	LoRA 20k	Δ (20k)
homoglyph	char	11.3%	12.1%	12.5%	17.2%	+5.9 pp
zwsp	char	41.9%	54.6%	55.2%	67.5%	+25.6 pp
whitespace	char	78.3%	86.0%	86.4%	89.8%	+11.5 pp
abbreviation	domain	10.2%	9.6%	7.3%	10.7%	+0.5 pp
honorific	domain	9.1%	6.6%	7.1%	11.0%	+1.9 pp
field_format	domain	16.0%	18.7%	16.8%	20.5%	+4.5 pp

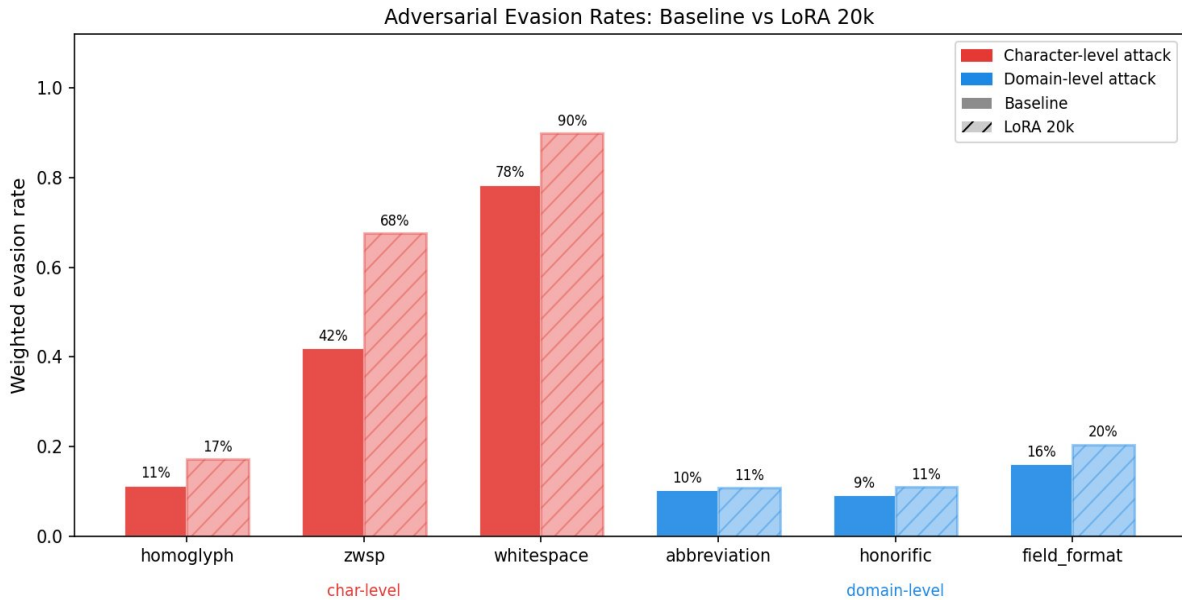


Figure 1. Evasion rates for all six attacks, baseline vs. LoRA 20k. Character-level attacks (left group) increase substantially and monotonically; domain-level attacks (right group) show negligible and inconsistent change.

Finding 1: LoRA provides near-zero defense against domain-level attacks. The original hypothesis predicted that fine-tuning on clinical prose would reduce evasion rates for domain-level attacks, since these attacks exploit the model's unfamiliarity with clinical formatting. In practice, domain-level evasion rates change by only ± 0.5 –4.5 pp across all LoRA configurations, a negligible and inconsistent shift with no monotonic trend. The baseline model may already have encountered clinical-adjacent text during pretraining, leaving little room for improvement.

Finding 2: LoRA increases character-level attack evasion; we term this *adversarial robustness forgetting*. ZWSP injection evasion rises from 41.9% to 67.5% (+25.6 pp) at LoRA 20k; thin-space injection rises from 78.3% to 89.8% (+11.5 pp); homoglyph substitution rises from 11.3% to 17.2% (+5.9 pp). All three character-level evasion rates increase *monotonically* with training set size. This pattern is analogous to catastrophic forgetting: just as a model fine-tuned on new data can lose performance on previously learned distributions, a model fine-tuned on clean in-domain data can lose the adversarial invariance it previously exhibited. Crucially, this robustness forgetting is invisible to standard utility metrics: the fine-tuned model achieves *higher* clean-input F1 on both domains, yet is *more* susceptible to character-level perturbations.

A plausible explanation for the asymmetry between attack types: clinical fine-tuning sharpens the model's sensitivity to specific token sequences found in clean clinical text. An entity the fine-tuned model detects with higher confidence under clean conditions may exhibit a sharper, more peaked probability distribution over entity labels. A character-level perturbation that changes even a single token ID then produces a larger departure from this narrower distribution, making detection more brittle. Domain-level attacks, which add context tokens rather than altering the entity's own token sequence, do not trigger this mechanism. This hypothesis is testable: examining per-token softmax entropy distributions before and after fine-tuning on clean inputs should show reduced entropy for entity-bearing tokens, a direction left for future work.

The tradeoff. Fine-tuning offers a genuine domain utility gain (clinical F1 +7.6 pp, no general-domain forgetting) and essentially no change in domain-level adversarial robustness. The cost is a significant and training-data-scaling increase in character-level evasion, a tradeoff that is attack-type-specific and invisible to aggregate robustness metrics.

4.4 LoRA Fine-Tuning Results

Clinical F1 (PII-injected MIMIC test set, 1,000 notes, 8,105 gold spans).

Model	Precision	Recall	F1
Baseline	0.467	0.820	0.595
LoRA 1k	0.505	0.975	0.666
LoRA 5k	0.507	0.988	0.670
LoRA 20k	0.508	0.990	0.671

LoRA fine-tuning delivers a clear and consistent improvement: F1 rises from 0.595 to 0.666–0.671 (+7.1–7.6 pp). The gain is driven almost entirely by recall: the baseline misses 27.6% of clinical person names and 32.3% of account numbers; with 1k fine-tuning examples, person recall rises to 95.1% and account number recall reaches 100%. Precision is essentially unchanged (~0.50). Performance saturates at 1k training examples: 5k and 20k add less than 0.5 pp.

Per-entity breakdown on the clinical test set:

Entity Type	Baseline R	LoRA 1k R	LoRA 20k R
<code>private_person</code>	72.4%	95.1%	98.0%
<code>private_date</code>	99.9%	100.0%	100.0%
<code>account_number</code>	67.7%	100.0%	100.0%

General-domain utility (AI4Privacy F1, corrected post-hoc evaluation).

Configuration	AI4Privacy Micro F1
Baseline	0.396
LoRA 1k	0.432
LoRA 5k	0.426
LoRA 20k	0.417

Attention-only LoRA does not cause catastrophic forgetting. General-domain F1 remains above baseline for all three configurations (+0.021–0.036). The monotonic decrease from LoRA 1k to LoRA 20k is small (0.015 pp over the full ablation range); all configurations remain above baseline utility (see Figure 3).

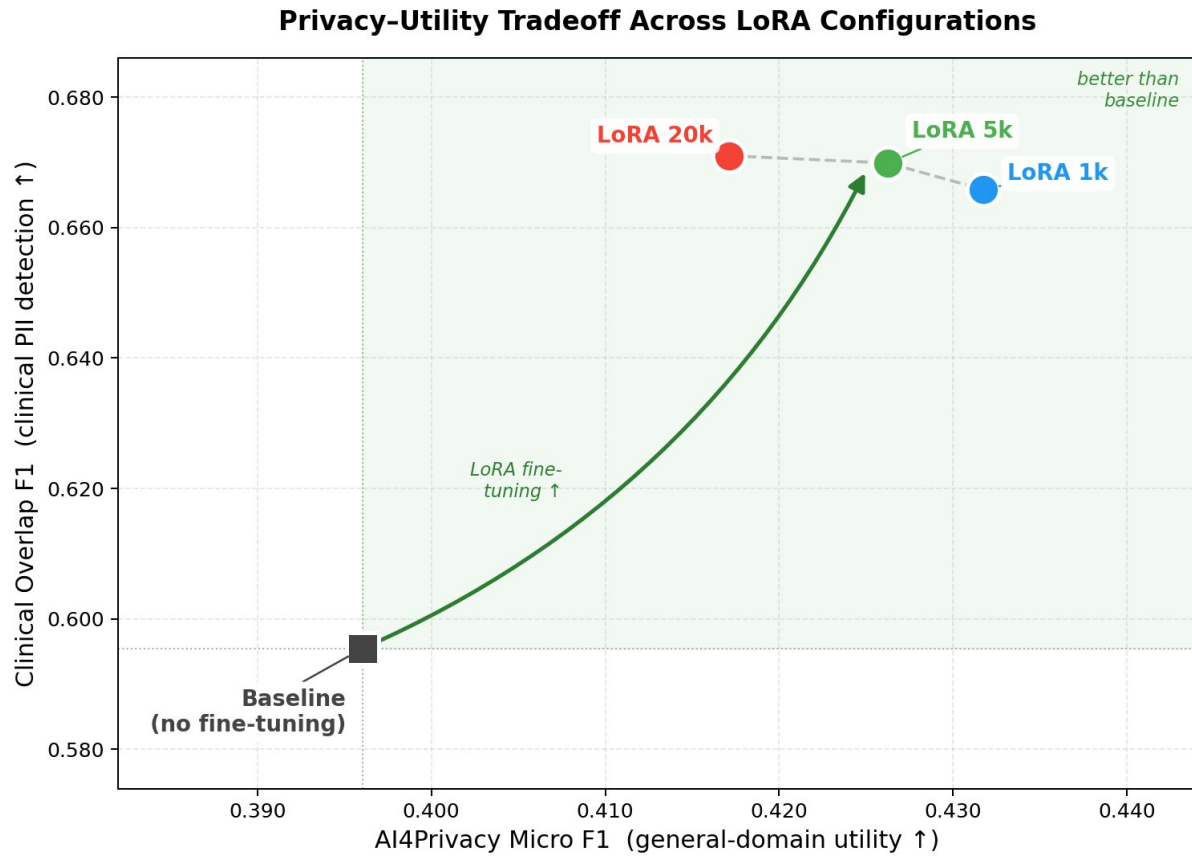


Figure 2. Privacy-utility tradeoff: general-domain F1 (x-axis, AI4Privacy) vs. clinical F1 (y-axis, PII-injected MIMIC). All LoRA configurations shift into the upper-right quadrant, reflecting improved clinical detection without general-domain utility loss.

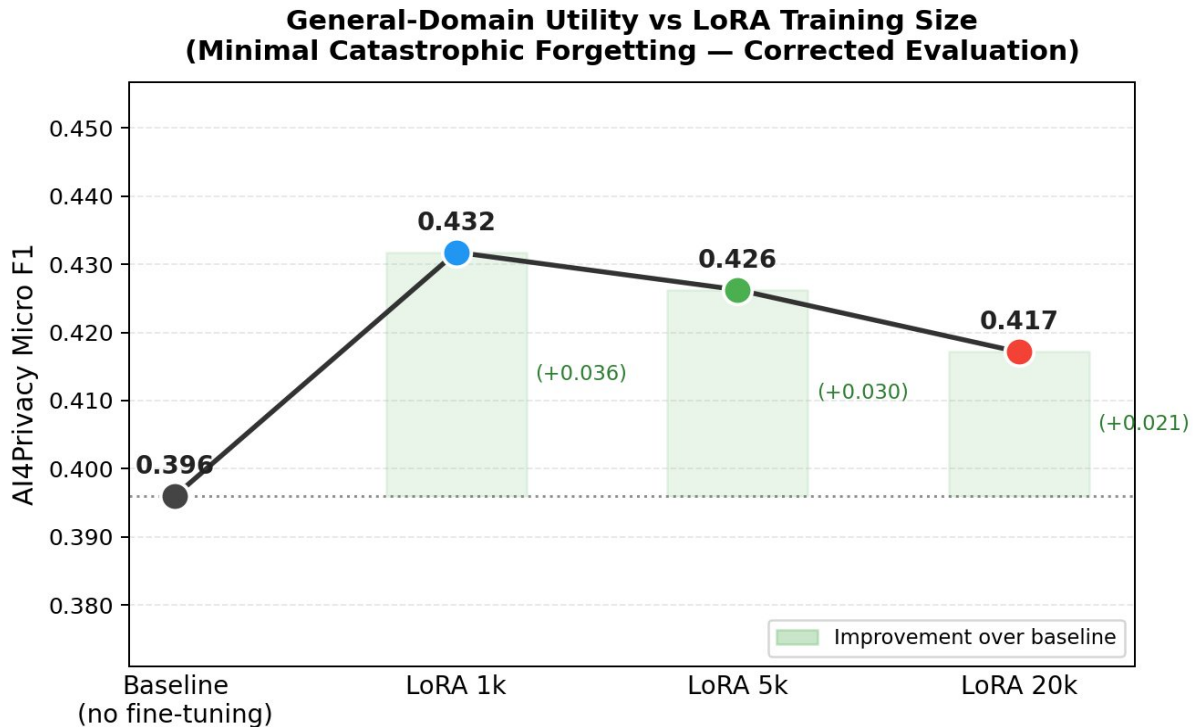


Figure 3. General-domain AI4Privacy F1 vs. LoRA training size (corrected post-hoc values). F1 remains above baseline for all configurations; no catastrophic forgetting is observed.

5. Conclusion

This project evaluated adversarial evasion of a large-scale PII detection system across general-domain and clinical text, and measured the effect of LoRA domain-adaptive fine-tuning on both detection utility and adversarial robustness.

The main findings are:

1. **Distribution shift degrades clinical detection.** The baseline model detects only ~1.06 entities per discharge note and misses 27–32% of injected clinical PII. Its behavior on clinical text is qualitatively different from general-domain.
2. **LoRA fine-tuning effectively closes the clinical detection gap.** Attention-only LoRA at rank 8 with as few as 1,000 training examples raises clinical overlap F1 from 0.595 to 0.666, closes recall gaps on `private_person` and `account_number`, and preserves general-domain utility.
3. **Domain-adaptive fine-tuning causes adversarial robustness forgetting for character-level attacks.** Domain-level evasion rates are essentially unchanged (± 0.5 – 4.5 pp), while character-level evasion rates increase substantially and monotonically with training data size (+5.9–25.6 pp), a tradeoff invisible to clean-input utility metrics.

Implications. Sub-tokenizer attacks (Unicode character manipulation) represent a persistent and exacerbated vulnerability for domain-adapted PII detection systems. They require no model access, are invisible to human reviewers, and worsen with fine-tuning. A defense would need to operate at the preprocessing layer; Unicode normalization (e.g., `unicodedata.normalize('NFKC', text)`) applied before inference would collapse many homoglyph and invisible-character variants to their canonical forms, eliminating the sub-tokenizer attack surface without modifying the model.

Limitations. All results use overlap F1 as the sole span-level metric; exact F1 is omitted because the systematic one-character tokenizer offset described in Section 3.5 would suppress exact match scores to near zero across all configurations, making them uninformative. The LoRA ablation covers only rank 8 and attention-only adaptation; results may differ for other ranks or expert-layer LoRA.

Future work. The most direct extension is an empirical evaluation of NFKC Unicode normalization as a preprocessing defense: applying `unicodedata.normalize('NFKC', text)` before inference should collapse homoglyph and invisible-character variants to their canonical forms, eliminating the sub-tokenizer attack surface without retraining. A secondary extension is a full per-entity breakdown of post-fine-tuning evasion rates across all 6 attacks and 3 LoRA configurations, which would clarify whether robustness forgetting is uniform across entity types or concentrated in specific categories.

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